ACETAZOLAMIDE

Sulfonamide
Metabolic acidosis

CHEMICAL NAME N-(5-sulfamoyl-1,3,4-thiadiazol-2-yl)acetamide
CLASS Carbonic anhydrase inhibitor
TRADE NAMES Diamox · Diamox SR · Acetazolamide sodium injection

THERAPEUTIC USES / INDICATIONS

- Glaucoma: acute angle-closure (pre-operative), open-angle (short-term adjunct).
- Acute mountain sickness (prevention and treatment).
- Idiopathic intracranial hypertension.
- Epilepsy (adjunct); metabolic alkalosis; periodic paralysis.

ADMINISTRATION

- **PO** tablets or SR capsules.
- **IV** injection (preferred over IM — painful, alkaline).
- Ensure adequate fluid intake.

DOSAGES

- **Acute glaucoma:** 500 mg IV/PO, then 250 mg every 6 h.
- **Glaucoma (chronic):** 250 mg 1–4 times daily.
- **Mountain sickness prevention:** 125 mg twice daily, start 1 day before ascent.
- **Idiopathic intracranial hypertension (off-label):** titrated by neurology; trial doses reached 4 g/day, but many stop at ~1.5 g/day due to side effects.

PHARMACOKINETICS

- **Onset:** PO 1 h; IV 2 min. **Duration:** 8–12 h.
- Not metabolised.
- Excreted unchanged in urine. **t½:** 3–6 h.

PHARMACODYNAMICS

- Inhibits **carbonic anhydrase** → ↓ bicarbonate reabsorption in proximal tubule (alkaline diuresis).
- ↓ aqueous humour production → ↓ intraocular pressure.
- ↓ CSF production; metabolic acidosis stimulates ventilation (altitude).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Paraesthesia (tingling fingers, toes, lips).
- Altered taste (carbonated drinks taste flat), polyuria.
- Fatigue, nausea, drowsiness.

ADVERSE EFFECTS (SERIOUS)

- Metabolic acidosis, hypokalaemia.
- Kidney stones (calcium phosphate).
- Blood dyscrasias, SJS (sulfonamide).
- Confusion in elderly/renal impairment.

CONTRAINDICATIONS & CAUTIONS

- Sulfonamide hypersensitivity.
- Hypokalaemia, hyponatraemia, hyperchloraemic acidosis.
- Severe hepatic impairment (hepatic encephalopathy); adrenal insufficiency.
- Long-term use in chronic angle-closure glaucoma.

KEY INTERACTIONS

- **High-dose aspirin:** severe acidosis, toxicity.
- **Phenytoin:** ↑ levels, osteomalacia.
- **Lithium:** ↑ excretion.
- **Corticosteroids:** hypokalaemia.

NURSING MANAGEMENT

- Monitor U&E, bicarbonate, CBC.
- Encourage fluids (stone prevention).
- Measure intraocular pressure/visual symptoms in glaucoma.
- Assess paraesthesia, confusion.
- Check sulfonamide allergy.

PATIENT EDUCATION

- Tingling of fingers and lips is common.
- Drink plenty of fluids.
- Altitude: still ascend slowly.
- Report rash, sore throat, flank pain.

KEY POINTS

- Carbonic anhydrase inhibitor.
- Paraesthesia, metabolic acidosis, stones.
- Glaucoma, altitude sickness, IIH.
- Sulfonamide — check allergy.

SOURCES CHECKED [337] Acetazolamide 500mg Powder for solution for injection (UK SmPC) [338] Hoffmann et al., J Headache Pain 2018
Full references p. 432–457